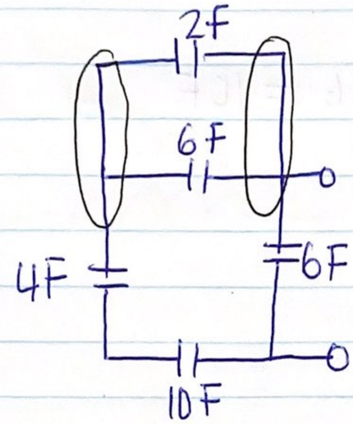
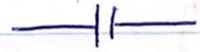
Chapter 6: Capicitance and Inductance



C = Capacitor (F)



↳ SI unit

L = Inductor (H)

↳ SI unit

Note: When capacitors are in series, use the parallel method to add them

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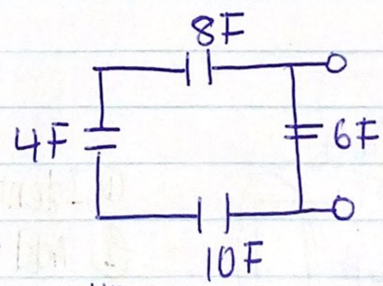
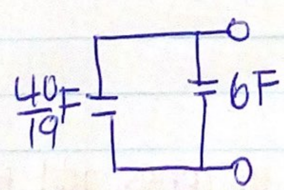
Using the circuit above, Calculate equivalent Capacitance. (C_{eq})

$2+6=8F \Rightarrow$

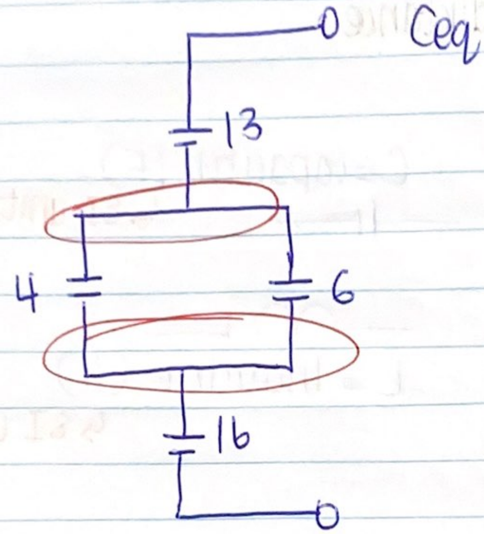


$\Rightarrow \left(\frac{1}{8}+\frac{1}{4}+\frac{1}{10}\right)^{-1} = \frac{40}{19}\,F$

$\Rightarrow$

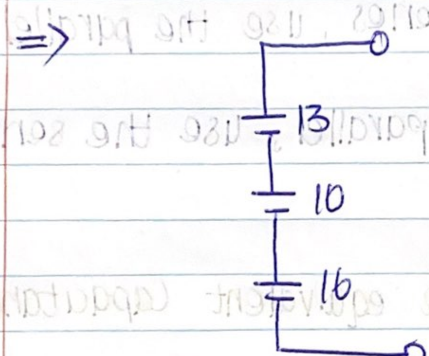


$\Rightarrow \frac{40}{19}+6 = \frac{154}{9}\,F$



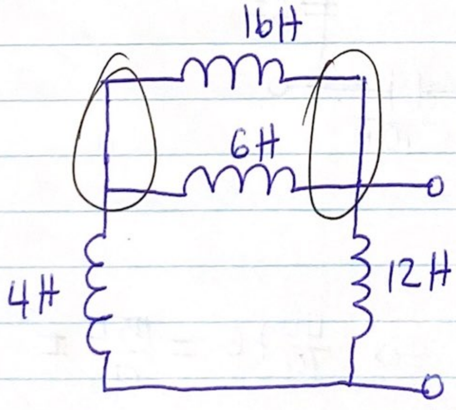
$4+6 = 10F$

$\Rightarrow$



$$\left(\frac{1}{13}+\frac{1}{10}+\frac{1}{16}\right)^{-1} =$$

Inductors



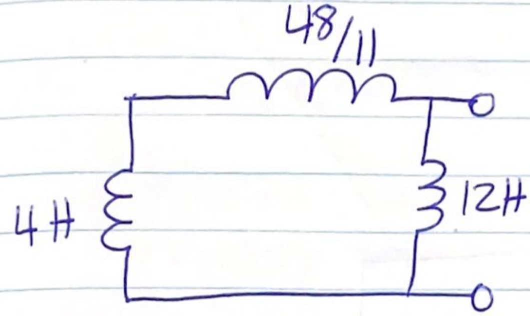
① Identify node

② Add in series or parallel

Note : When inductors are in series, add them in series
When inductors are in parallel, add them in parallel

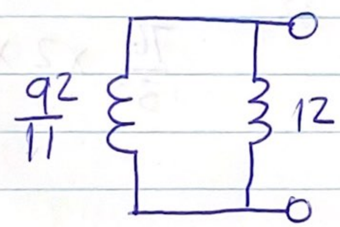
Calculate Ieq

$$\left(\frac{1}{16}+\frac{1}{6}\right)^{-1} = \frac{48}{11}$$



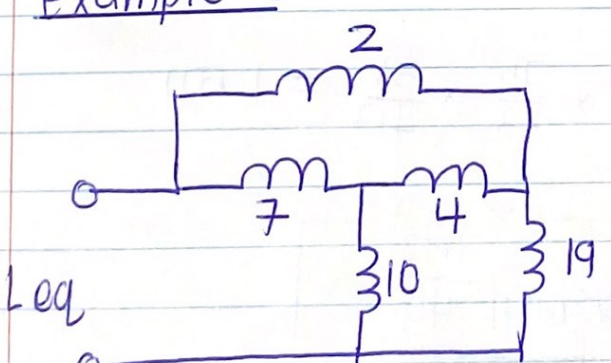
$$\frac{48}{11}+4 = \frac{92}{11}$$

$\Rightarrow$

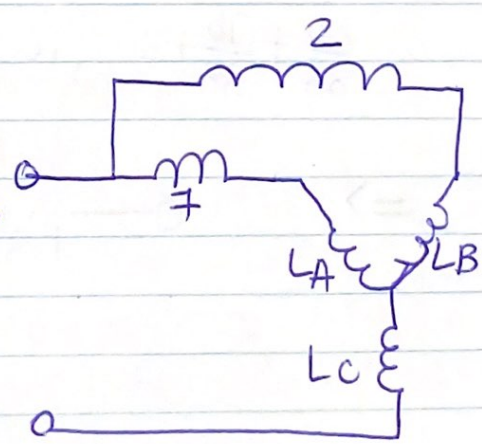


$$\frac{92}{11}+12 = \frac{224}{11}$$

Example 2



$\Rightarrow$

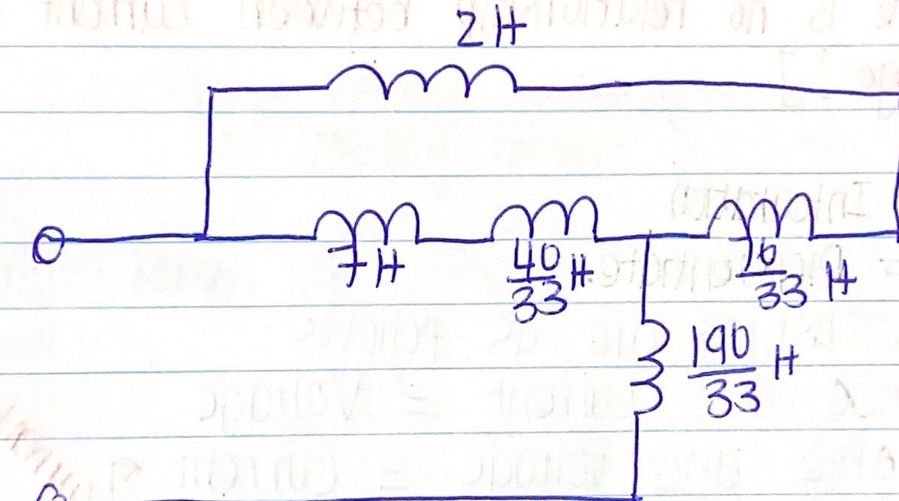


$$L_A = \frac{4\times10}{4+10+19} = \frac{40}{33}$$

$$L_B = \frac{4\times19}{4+10+19} = \frac{76}{33}$$

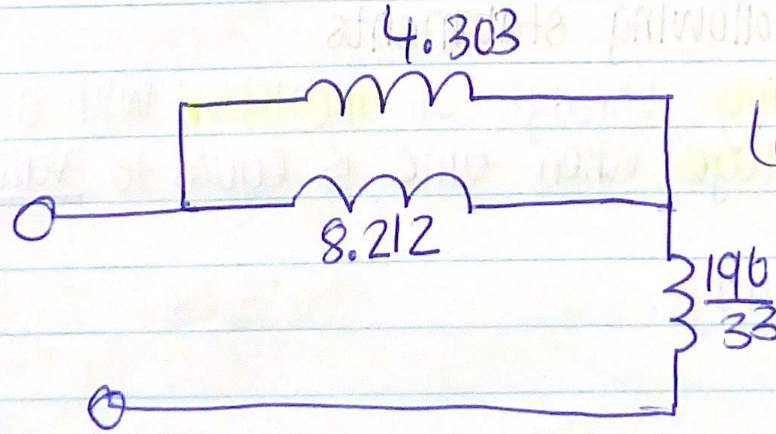
$$L_C = \frac{10\times19}{4+10+19} = \frac{190}{33}$$

⇒



$\cancel{70}$ $7+\frac{40}{33} = \overset{B}{8.212}$ | $\frac{76}{33}+2 = \overset{A}{4.303}$

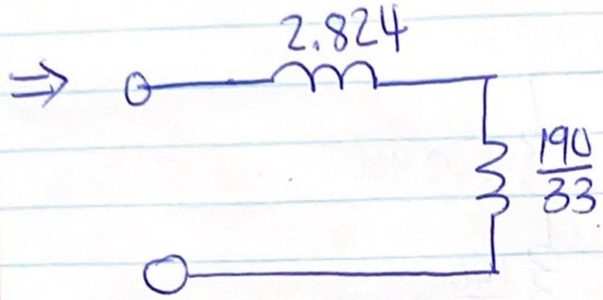
⇒



$$\left(\frac{1}{4303}+\frac{1}{8.212}\right)^{-1}$$

$$= 2.824\ H$$

$\Rightarrow$



$$2.824 + \frac{190}{33} = 8.58 \text{ H}.$$

enemies enemies

C I V L

Capacitance Current Voltage Inductance

friends friends

[NB: There is no relationship between current and Voltage!]

Friends = Integration
Enemies = Differentiate.
The relationships are as follows:

Voltage
1. Capacitance and Current = Voltage
2. Capacitance and Voltage = Current
3 Voltage and Inductor = Current
4 Current and Inductor = Voltage.

current

Given the following statements
① Current passing through an inductor with a given value. Calculate voltage when time is equal to value.
Given.
$\left.\begin{matrix} I \\ L \end{matrix}\right\}$ find Voltage.

Voltage with respect to or as a function of time

$\therefore V(t) = L \frac{di(t)}{dt}$ → diff current / → time

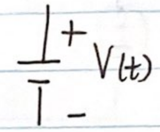
② voltage across a capacitor

Given

V

C

I = ?

across means 

$\therefore I(t) = C \frac{dV_t}{dt}$ → diff Voltage / → time

③ Current passing through a capacitor between t_1 and t_2

Given:

I

C

V = ?

$\therefore V(t) = \left[\frac{1}{C}\int_{t_1}^{t_2} i(t)\, dt\right] + V(0)$

$V(0)$ = voltage when time is zero, value is usually given.

④ Voltage across an inductor between t_1 and t_2

Given

V

L

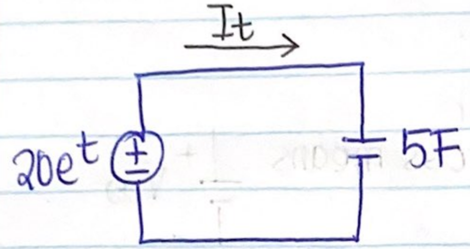
I = ?

$\therefore I(t) = \left[\frac{1}{L}\int_{t_1}^{t_2} V(t)\, dt\right] + I(0)$

$I(0)$ = Current when time is zero, value usually given.

Examples

1.



Calculate It.

Given

V and C

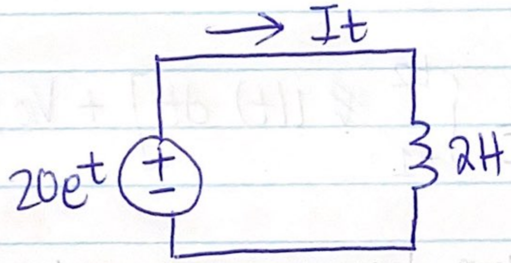
$$\Rightarrow I(t) = C\frac{dV_t}{dt}$$

time =

Use calculator.

$I(t) = 164,87\ A$

2.



Calculate It

Given

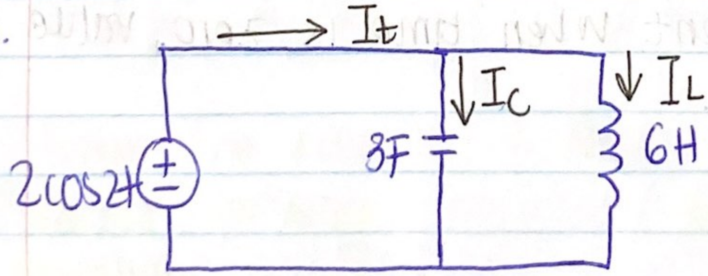
V and L

$$I(t) = \left[\frac{1}{L}\int_{t_0}^{t_1} V(t)\,dt\right] + I(t)$$

$$= \left(\frac{1}{2}\int_{0}^{0.5} 20e^{x}\right] + 0 \quad \Big\}\ \text{use calculator}$$

$I(t) = 6,487\ A$

3.



NB: set calculator to radians.

Shift → Mode → 4

$I_t = I_C + I_L$ $\qquad$ $I_0 = 5A$

In = Out

Given

V and C $\Rightarrow$ find I_C

$$I(C) = C \frac{dV_t}{dt}$$

$$\text{Calculator} = 3\left(\frac{d}{dx}\, 2\cos(2x)\right)\Big|_{x=2}$$

$$I_C = \underline{9.08\,A}$$

Given

V and L $\Rightarrow$ find I_L.

$$I_L = \left(\frac{1}{L}\int_{t_0}^{t_1} V(t)\,dt\right) + I(0)$$

$$\text{Calculator} \Rightarrow \quad = \left(\frac{1}{6}\int_0^2 (2\cos(2x)\right) + 5$$

$$= \underline{4.87}.$$

$$\therefore \quad I(t) = I_C + I_L$$

$$= 9.08 + 4.87$$

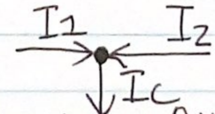
$$= \underline{13.96\,A}$$

<u>Example 4</u>

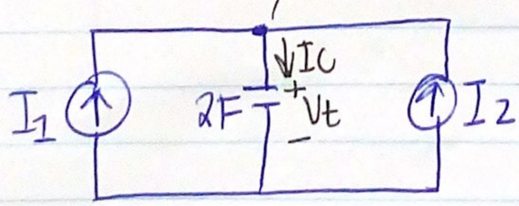
Find $V_{C(t)}$ at $t = 2\text{sec}$, $V(0) = 0$

$i_1 = 2e^{-4t}$

$i_2 = 8t$



In = Out.



Given

I and C

$\overline{Vt = ?}$

$I_c = I_1 + I_2$

$I_c = 2e^{-4t} + 8t.$

$$V_c(t) = \left[\frac{1}{C}\int_{t_0}^{t_1} I(t)\,dt\right] + V_0$$

$$= \left(\frac{1}{2}\int_{0}^{2} (2e^{-4t} + 8t)\right] + 0$$

$V_c(t) = 8.25\,V$